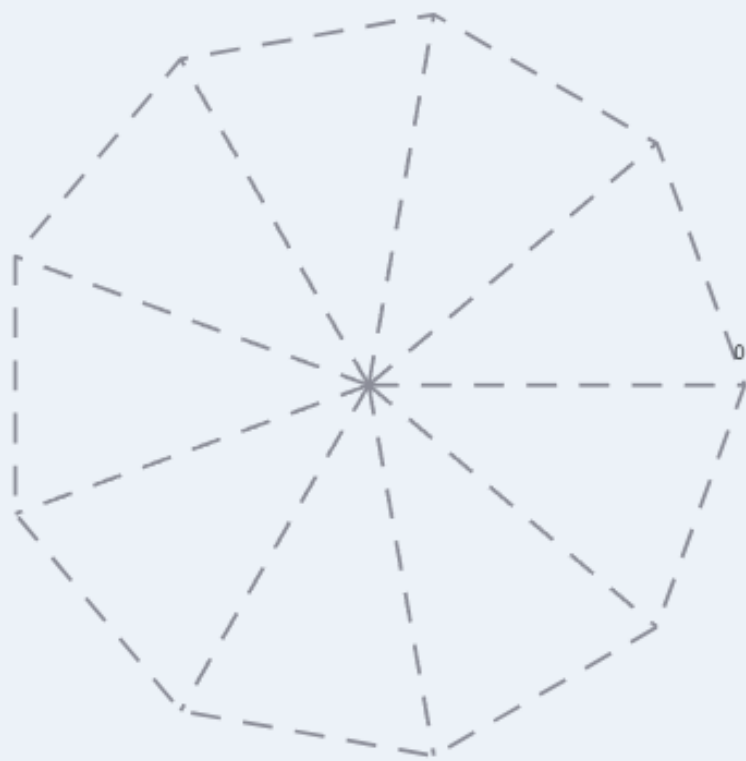




Choose at most three topics that you find most difficult (multiple vote on a single choice is allowed; max 3 votes)



- A 0% Find Cartesian from Parametric Equations
- B 0% Find Tangent Lines to Curve
- C 0% Implicitly defined parametrizations
- D 0% Area enclosed by curve
- E 0% Length of Curves
- F 0% Graph the set of polar coordinates points
- G 0% Polar to Cartesian Equations
- H 0% Cartesian to Polar Equations
- I 0% Nothing is difficult





We want to find the area under $y = x^3$ over $[0, 1]$

. Choose the correct expression for the area using the parametrization $x = t^2$, $y = t^6$, $0 \leq t \leq 1$.

☐
$$\int_0^1 2t^6 dt$$

☐
$$\int_0^1 2t^7 dt$$

☐
$$\int_0^1 6t^6 dt$$

☐
$$\int_0^1 6t^7 dt$$

 Responses Hidden



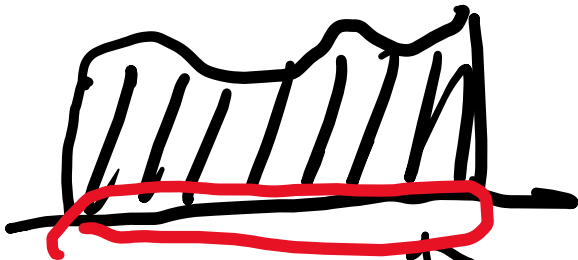
Ex 10.2 Q24 (a)

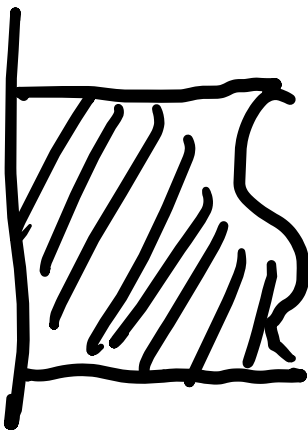
$$y = x^3 \quad \text{over } [0, 1] \Rightarrow 0 \leq x \leq 1$$

$$x = t^2, \quad y = t^6 \quad [\text{what is } t]$$

$$0 \leq t \leq 1 \Rightarrow 0 \leq t^2 \leq 1$$
$$\Rightarrow \underline{0 \leq x \leq 1}$$

Two types of areas:

① $\int y \, dx \rightarrow$ 

② $\int x \, dy \rightarrow$ 

$y = x^3$



$\Rightarrow \text{Area} = \int y \, dx$

$$\text{Area} = \int y \, \underline{dx} \quad (\text{Draft})$$

$$\text{Area} = \int_0^1 t^6 - 2t \, dt$$

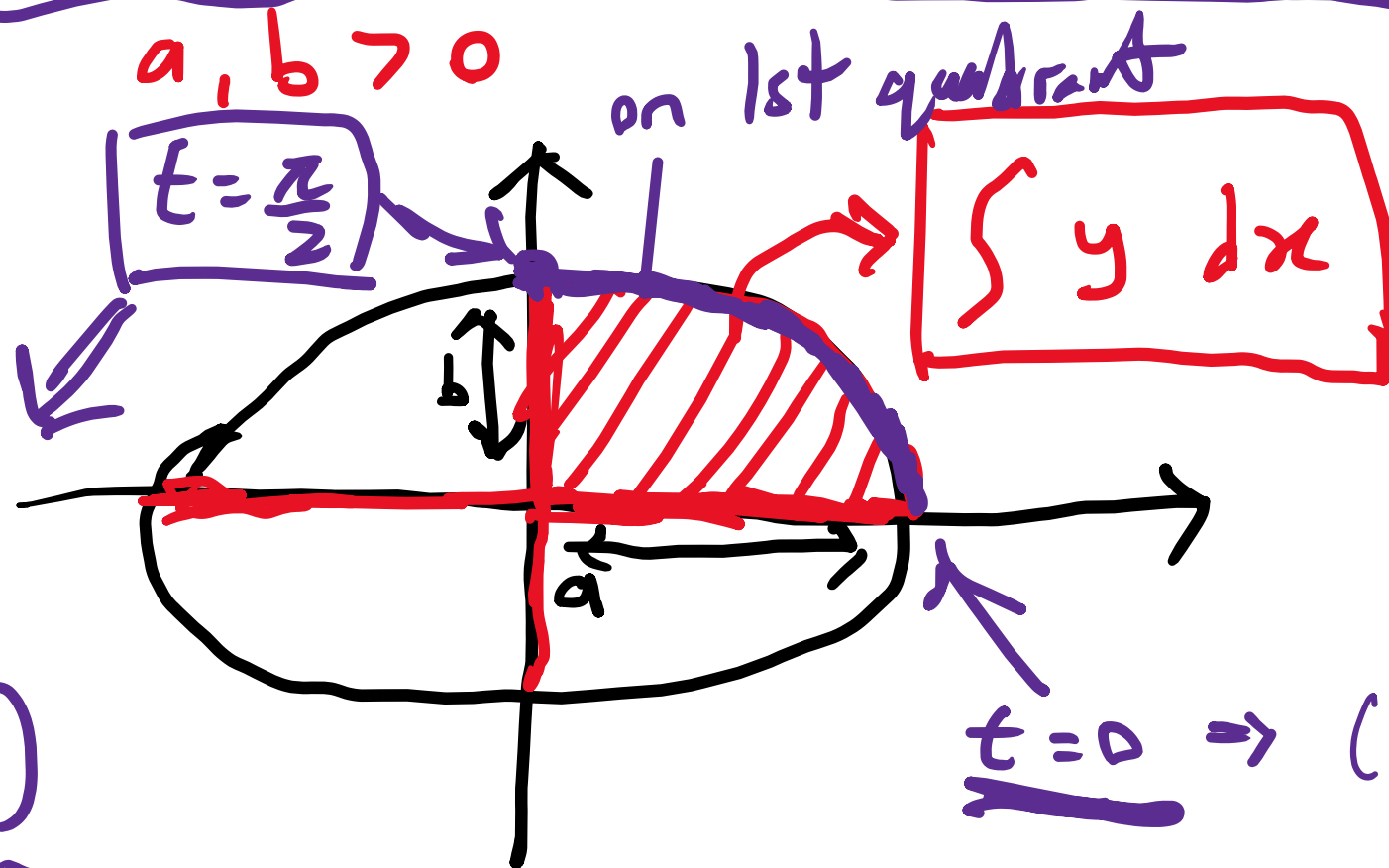
$$x = t^2 \Rightarrow dx = \underline{2t \, dt}$$

Q23

$$x = a \cos t, \quad y = b \sin t, \quad 0 \leq t \leq 2\pi$$

$$a, b > 0$$

Ellipse



$$\int y \, dx \quad \text{or} \quad \int x \, dy$$

(x, y)

$(0, b)$

$$a \cos \frac{\pi}{2} \quad b \sin \frac{\pi}{2}$$

$$\underline{t = 0} \Rightarrow (x, y) = (a, 0)$$

$$\text{Area} = 4 \left(\text{shaded region} \right)$$

$$x = a \cos t$$

$$\Rightarrow dx = -a \sin t$$

$$\int_{t=0}^{t=\frac{\pi}{2}} y \, dx$$

$$\text{Area} = 4 \int_0^{\frac{\pi}{2}} b \sin t (-a \sin t) \, dt$$

$$= \dots$$

End  Start

Starting point
at (x_0, y_0)

Find t_1 s.t.
 $x(t_1) = x_1$
Endy point
at (x_1, y_1)

$y(t_1) = y_1$

Find t_0 so that $x(t_0) = x_0$
 $y(t_0) = y_0$

$$A = \int_{t_0}^{t_1} \underline{y} \, \underline{dx} \quad (\text{use sub.})$$



Which polar coordinate pairs label the same point?

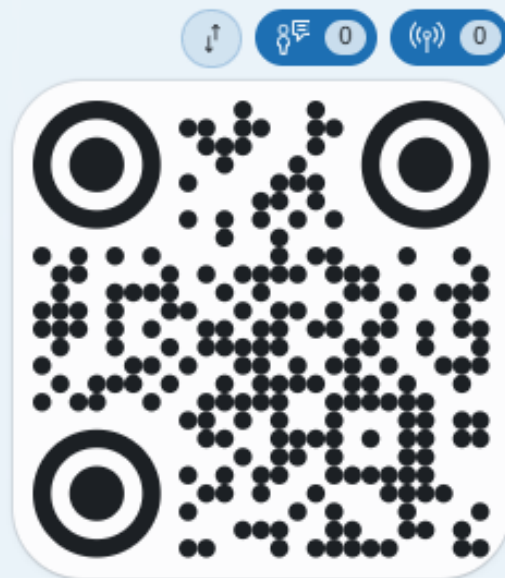
$(2, -\frac{\pi}{3})$ and $(-2, \frac{\pi}{3})$

$(-2, \frac{\pi}{3})$ and $(2, -\frac{2\pi}{3})$

(r, θ) and $(r, \theta + \pi)$

$(-r, \theta)$ and $(-r, \theta + \pi)$

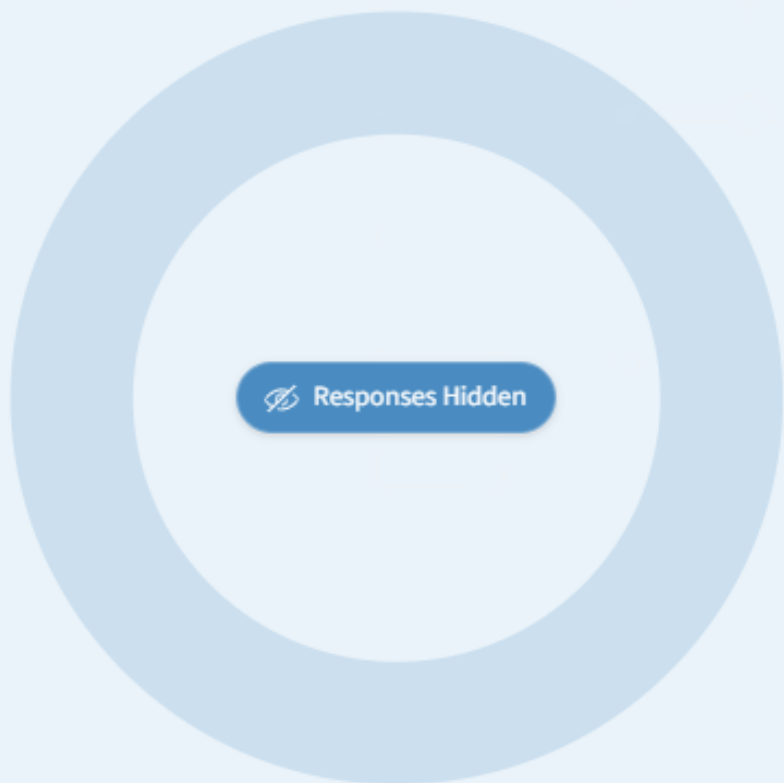
Responses Hidden





Choose the correct graph for $0 \leq \theta \leq \pi/2$, $1 \leq |r| \leq 2$

0 0



A

0%



B

0%



C

0%



D

0%



Ex. 10.3 Q26

My own method:

- ① Look at Θ $\rightarrow$ Plot that region
- ② Identify regions corresponding to $r > 0$
and $r < 0$
- ③ Plot (find the intersection between
① and circles drawn based on
the conditions on r)

① Look at θ

$$0 \leq \theta \leq \pi/2$$

• $\theta = 0$

• $\theta = \pi/2$

$$1 \leq |r| \leq 2$$

$\Downarrow$

$$1 \leq r \leq 2 \text{ or } -2 \leq r \leq -1$$

$r \geq 0$

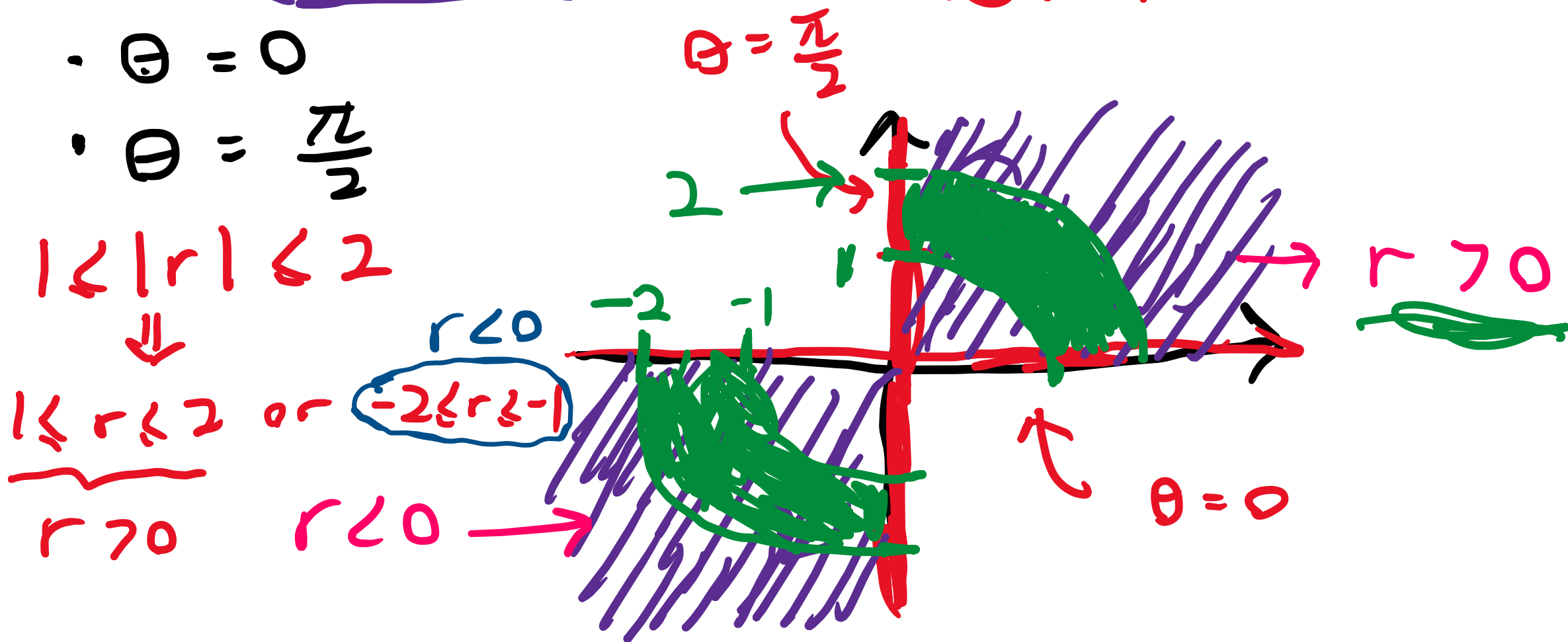
$r \geq 0$

$r < 0$

$r < 0$

② Identify regions for
 $r \geq 0, r < 0$

③ Plot



Ex. 10.2 Q20

$$t = \ln(x-1), \quad y = te^t, \quad t=0$$

Aim: Find the slope of the curve at $t=0$

$$\frac{dy}{dx} \Big|_{t=0} \rightarrow \frac{dy}{dx} = \frac{\boxed{dy/dt}}{\boxed{dx/dt}}$$

x, y are func. of t

$$\left[\frac{d}{dt}(x) = x' \text{ or } \frac{dx}{dt} \right] \quad \frac{d}{dt} x = 0$$

$$\frac{dx}{dt} : t = \ln(x - t) \Rightarrow \begin{array}{l} \text{Inner : } x - t \\ \text{Outer : } \ln \underline{u} \end{array}$$

$$\Rightarrow 1 = \frac{1}{x - t} \left(\frac{dx}{dt} - 1 \right)$$

$$\Rightarrow \frac{dx}{dt} = x - t + 1$$

$$\frac{dy}{dt} : y = t e^t \Rightarrow \frac{dy}{dt} = t e^t + e^t$$

$$\therefore \frac{dy}{dx} = \frac{dy/dt}{dx/dt} = \frac{t e^t + e^t}{\boxed{x} - t + 1}$$

$$\text{When } t=0, \quad t = \ln(x-t)$$

$$\Rightarrow 0 = \ln(x-0)$$

$$\frac{e^0}{\Rightarrow x=1} = e^{\ln x}$$

$$\therefore \left. \frac{dy}{dx} \right|_{t=0} = \dots$$



When converting cartesian equation $(x - 5)^2 + y^2 = 25$ to polar equation, I write

$$x^2 - 10x + 25 + y^2 = 25 \implies x^2 + y^2 = 10x \implies r^2 = 10r \cos \theta \implies r = \pm \sqrt{10r \cos \theta}$$

. What do you think about this argument?

$$r^2 = x^2 + y^2$$

Sounds good to me.

It is flawed.

It is flawed but I cannot find a reason for that.

It is technically correct but it gives no information on the values of

r .



Responses Hidden

Edit

Show Choices



Show Responses



Lock



Show Correctness



$$r^2 = 10r \cos \theta$$

$\Rightarrow$

$$r = 10 \cos \theta$$

(divide by r)

Ignore them

represent points (r, θ) that satisfy

because $r \neq 0$

$$r = 10 \cos \theta$$

$$r = 0 \Rightarrow \text{Origin} = \underline{(0, 0)}$$

$$(x - 5)^2 + y^2 = 25 \Rightarrow (0, 0) \text{ satisfies this eq.}$$